Over-representation analysis

Which biological terms appear in a list of features more often than chance would put them there.

AT A GLANCE

Where	Statistics → ORA, titled GO Enrichment (ORA)
Data levels	Protein and phosphoprotein. Phosphosite level is refused: a site list is not a gene list.
Needs	Gene names in the dataset, and an upstream result — a volcano comparison or a selected ANOVA cluster.
Organism	Human (*Homo sapiens*), the only bundled Gene Ontology database. Nothing falls back to it: another organism is refused rather than silently answered.
The test	One-sided hypergeometric test per term, Benjamini–Hochberg corrected across the terms that overlap the query.
Output	One dot plot and one table per gene list — up and down from a volcano, one from an ANOVA cluster — plus the gene lists themselves as text.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the page answers

A comparison returns a few hundred genes, and the question is whether that list is about anything: whether its genes share an annotation more often than a list of the same length drawn at random would.

Over-representation answers it by counting. For one Gene Ontology term, four numbers suffice: how many genes carry the term, how many are in the query list, how many are in both, and how large the universe all three are drawn from is. The hypergeometric distribution gives the probability of an overlap at least that large by chance — the term's p-value — and the procedure is repeated for every term of a usable size and corrected across them. The fourth number decides the answer, and it is the one the page makes you choose.

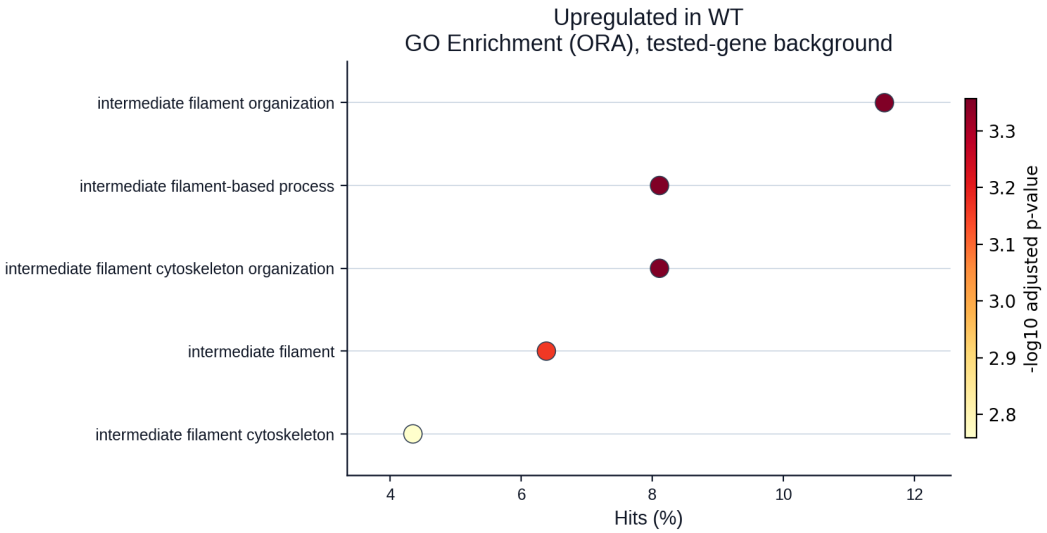


Figure 1.1. The dot plot for the upregulated gene list of WT against GRK, at protein level and at the default settings. Five terms survived the cutoff, all of them about intermediate filaments; the x axis is Hits (%), the colour is $-\log_{10}$ of the adjusted p-value with the most significant term at the top, and the dot area is the number of query genes the term contains.

Reading it

- One dot, one term.
- The x axis, Hits (%): the share of the term's genes that are in the query list — the overlap divided by the term's size within the background.
- The dot's area: the size of the overlap, in genes. It is relative rather than absolute — the figure carries no size key, and the exact count is the intersectionSize column of the table below — so read it as a comparison between terms in the same figure and nowhere else. A large dot far left is a large term barely covered; a small dot far right is a small term covered almost entirely.
- The colour and colourbar: $-\log_{10}$ of the adjusted p-value. Terms are ordered by it, most significant at the top.
- The title names the direction, the condition it belongs to and the background, so the figure carries all three with it.

> **THE DIRECTION IS ONE CONDITION'S**
>
> The title of the upper figure reads Upregulated in WT, and names no other con-

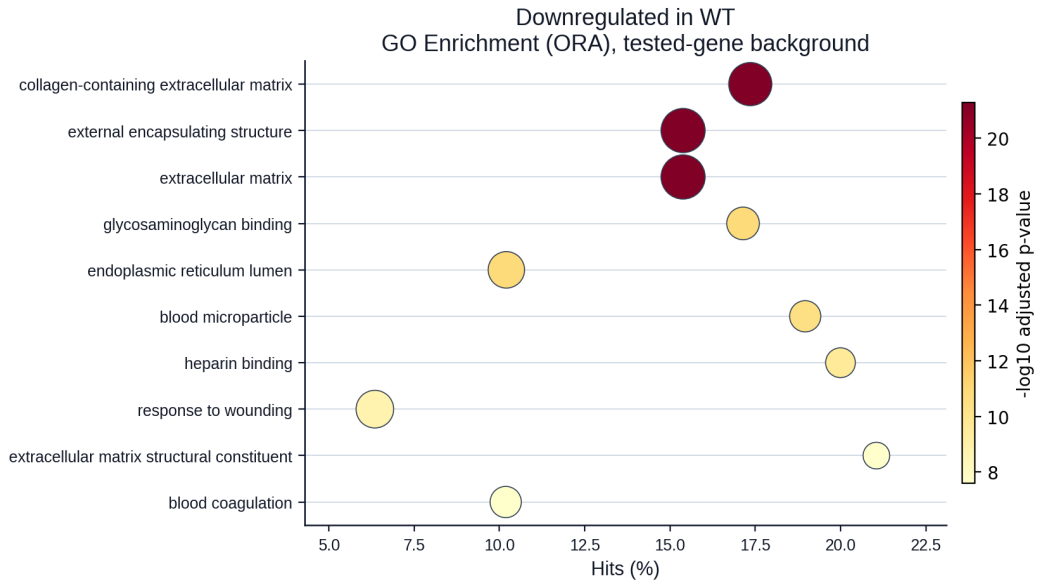


Figure 1.2. The downregulated list of the same comparison, which the page draws directly under the one above. Ten terms, all extracellular matrix and wound response. The colour scale is this term set's own — the two figures are separate analyses of separate gene lists, and neither their colours nor their dot sizes are comparable across the pair. What is comparable is what they are about.

dition. That is the whole statement: the upstream volcano computes $\log_2$ FC as mean(Condition 1) − mean(Condition 2), so a feature called Upregulated is higher in Condition 1, and the terms enriched among those features are the terms that are over-represented among the genes higher in Condition 1. Saying *and lower in GRK* as well would be the same fact a second time, and the second telling is what makes a reader stop and work out which way round the figure is.

EASY TO GET WRONG

The background is the set of genes the query could have been drawn from, and the two settings answer different questions. Tested genes in this analysis — the default — uses every gene that received the upstream test, so the result is about what moved among the things this experiment could see. All annotated human genes uses the whole Gene Ontology universe, and will report terms enriched only because their genes are detectable by mass spectrometry at all. Whichever is chosen is intersected with the annotated universe before anything is counted, and the mode and both sizes are printed above the table and written into the exported CSV.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Protein or phosphoprotein.	Protein	page
Analysis source	Where the gene list comes from: Volcano or ANOVA. The card below changes with it.	Volcano	page
Source	Volcano or Volcano Control under the volcano source; fixed at ANOVA Cluster under the ANOVA source.	Volcano	page
Data Source	The matrix the upstream test runs on.	the most processed matrix available	page
Condition 1, Condition 2	Volcano source. The comparison, in the same orientation as the volcano page: up means higher in Condition 1.	first condition in the annotation	page
Condition 1 Control, Condition 2 Control	Volcano Control only. Four distinct conditions are required.	first condition in the annotation	page
Test type, P-value threshold, log2 FC threshold, Use uncorrected p-values	The upstream volcano settings, as on the volcano page. They decide which features are called up or down, and so what the gene lists contain.	Unpaired (Student's t, equal variance), 0.05, 1, off	page
Clusters, Cluster method, Protein cluster	ANOVA source. The clustering to rebuild and which of its clusters supplies the gene list. All three must match the ANOVA page the cluster was chosen on, or the numbering differs.	4, Average (UPGMA), none	page
Organism	Which Gene Ontology database to match against. Human (Homo sapiens) is the only entry.	Human (Homo sapiens)	page
Adj p-value cutoff	Terms with an adjusted p-value above this are dropped from the result.	0.05	page
Top terms	How many surviving terms to keep, taken from the most significant end.	10	page
Min term size, Max term size	The size a term must fall between to be tested, counted *after* intersection with the background rather than in the database. Very small terms give unstable percentages, very large ones are too general to interpret.	20, 300	page
Background	Tested genes in this analysis (recommended) or All annotated human genes (global).	Tested genes in this analysis	page
Width, Height, DPI	The size of both figures, on screen and in the download alike — the two are the same picture here, unlike the pages where the download is rendered separately. They are a *minimum*: see the note below.	26 cm × 12 cm, 300	page

Controls on this page (continued)

Control	What it does	Default	Where
Download format	PNG, JPG, WEBP, SVG or PDF, beside each figure. One setting serves both figures, so a pair downloaded for the same document arrives in the same format.	PNG	page

> **WHY THE SIZE IS A MINIMUM**
>
> Gene Ontology descriptions can be too long for a compact figure. The page does three things about it, in this order: the term labels are wrapped to the strip of width beside the plot, the canvas is enlarged — beyond **Width** and **Height** — until every label is inside it, and only if the largest permitted canvas still cannot hold them is the label text set smaller than asked for. This improves label fitting but cannot guarantee a figure for invalid or empty input. Set the two numbers to the smallest figure you want and let it grow; setting them large does not make the terms any easier to read, it only spreads five dots over a page.

In the plot appearance panel

Control	What it does	Default	Where
Title, axis labels, ticks, font, frame, grid, background	As for every other plot; described in Part XI. The dot colours are the significance scale and are not a setting.	the plot's own	appearance

3 How it is computed

1. **The gene list.** The upstream analysis runs through the same code path the Volcano and ANOVA pages use. From a volcano the Upregulated features form one query list and the Downregulated features another, analysed separately; from an ANOVA the selected cluster forms a single list. Gene labels are split on semicolons where an entry names a group, trimmed and deduplicated.
2. **The background.** Tested genes takes every gene that received the upstream test, not only the significant ones; Global takes every gene carrying at least one term. Either is intersected with the annotated universe, and that intersection is the population the test draws from.
3. **The terms.** Every term is intersected with the background, and those whose remaining

size falls between **Min term size** and **Max term size** are eligible. The database covers the three Gene Ontology branches — biological process, molecular function, cellular component — tested together in one family.

4. **The test and the cut.** Each eligible term sharing at least one gene with the query is tested, the p-values are corrected, and the terms surviving **Adj p-value cutoff** are sorted and cut to **Top terms**.

THE HYPERGEOMETRIC TEST

For one term, with N the number of genes in the background, K the number of them that carry the term, n the number of query genes that are in the background and k the overlap, the p-value is the probability of drawing at least k of the term's genes in n draws without replacement:

$$p = \sum_{j=k}^{\min(K,n)} \frac{\binom{K}{j}\binom{N-K}{n-j}}{\binom{N}{n}}.$$

The test is one-sided: only over-representation is reported, never depletion. Hits (%) on the figure is $100\,k/K$.

CORRECTION

Benjamini–Hochberg, run over the terms that share at least one gene with the query. Zero-overlap terms are excluded from this implementation's correction family. This is a specific analysis convention, not a universal requirement of Benjamini–Hochberg: correcting over all size-eligible terms, including zero-overlap terms with $p = 1$, can give different adjusted values. Report the convention when comparing with another enrichment tool. The two counts are in the table: testedTermCount is how many terms passed the size filter, overlapTested the smaller number the correction actually used.

The two corrections on this page are separate and are reported separately. correctionMethod is always Benjamini-Hochberg and describes the term p-values; upstreamCorrectionMethod describes the volcano or ANOVA that chose the genes, and reads none (uncorrected) when **Use uncorrected p-values** was on. Each query list — up, down or the cluster — is corrected independently of the others.

Edge cases

- A query gene absent from the background is ignored; in Global mode a symbol absent from the database is dropped.
- An empty query list, or one with no eligible overlapping term, produces no figure and a message saying which.
- Where at least 50 distinct query symbols are supplied and under 20 % of them are in the

human database, a warning is shown — the practical sign that the data is from another organism, for which the result would be arithmetically valid and biologically meaningless.

- **Min term size** above **Max term size** is refused.

4 The result table

One table per gene list, holding the terms that survived the cutoff. Each row carries the term and, repeated on every row so a downloaded file can be read alone, the settings it was produced under.

Columns in this table

Column	What it holds
source	The Gene Ontology branch: BP, MF or CC.
termId	The GO identifier.
name	The term description, as shown on the figure.
termSize	Genes carrying the term *within the background*.
intersectionSize	Genes in both the term and the query.
hitPercent	100× intersectionSize / termSize — the x axis of the figure.
pValue	The raw hypergeometric p-value.
adjPValue	Benjamini–Hochberg adjusted across the overlapping terms of this query.
intersectingGenes	The overlap itself, named.
backgroundMode, backgroundSize, annotatedBackgroundSize	Which background was chosen, how many genes were submitted as it, and how many of those carry any annotation.
correctionMethod, upstreamCorrectionMethod	The correction on the term p-values, and the one used upstream to choose the genes.
testedTermCount, searchSpace, overlapTested, passingCutoff, displayed	How the search space narrowed: terms passing the size filter, terms actually tested, terms under the cutoff, terms kept by **Top terms**.
method, organism, analysisSource, datasetLevel, dataSource	What was run, against which database, from which upstream analysis, at which level and on which matrix. The upstream conditions, test type, thresholds and cluster settings follow on the same row.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
genes = different_genes(data, meta, "WT", "GRK", workflow="Protein")
fig = enrichment_analysis(genes["Upregulated"], background=genes["Background"])
```

R

```
genes <- different_genes(data, meta, "WT", "GRK", workflow = "Protein")
fig <- enrichment_analysis(genes$Upregulated, background = genes$Background)
```

The page hides the join between the two analyses; in a notebook it is explicit. different_genes runs the same volcano and hands back three gene lists — Upregulated, Downregulated and Background, the last being every gene that was tested — and each query list is then enriched on its own, so the figure above is one of two the page produces from the same call. The background is passed in by hand rather than chosen from a menu: background_mode stays at "experiment", which is the page's Tested genes in this analysis, and enrichment_data returns the same table the figure is drawn from when the numbers are wanted instead of the picture.

6 Exporting

Each figure has its own format selector and Download Plot button beside it, and its own Open Image link; each table has its own Download CSV. The downloaded figure is the one on screen, at the width, height and resolution set on the page — there is no separate render, so what is downloaded cannot differ from what was looked at. Choose PDF or SVG for a figure that will be scaled or edited, PNG for a slide. The file is named for its direction and its comparison, so ora_up_protein_WT_vs_GRK.png and its downregulated counterpart do not have to be told apart afterwards.

The gene lists download separately as plain text: the up list, the down list, and — in Tested genes mode — the background itself, which is what makes the result reproducible by somebody else.

The bundled Gene Ontology table is fixed for a release, so the result does not print a database identity: the release of the application is the provenance. Quote the version on this manual's title page when a method section needs one.

7 Further reading

Gene Ontology: enrichment analysis explains why the reference population matters. SciPy: false discovery control discusses multiple testing. CoPYlot's tested-term family is the specific convention documented above.

Behind this page

Function	What it produces
enrichment_data	The over-representation table for one gene list
enrichment_analysis	The dot plot of that table
load_go_db	The bundled Gene Ontology term-to-gene table